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DFROG-MAC: A Dynamic Fragmentation-Based MAC for Prioritised Emergency Data Management in Vehicular Networks

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ABSTRACT

The rapid advancements in vehicular ad hoc networks (VANETs) call for development of effective networking schemes. Managing heterogenous traffic in VANETs becomes a critical challenge, especially when dealing with critical scenarios. In this paper, we present a novel dynamic fragmentation-based MAC protocol, DFROG-MAC for Internet of Things (IoT) applications in VANET environment. This protocol is focused on facilitating prioritised heterogenous traffic in sensor networks and hence, can offer a distinguished quality of service for various application areas such as vehicular, industrial or body sensor networks. DFROG-MAC deploys fragmentation scheme for low priority data, so the high priority data may interrupt and access channel without needing to wait for the complete transmission of lower priority data. The fragment size for the lower priority data is dynamically adjusted at the runtime based on the frequency of urgent traffic arrival. This dynamic approach helps to ensure that the channel does not remain idle, and lower priority traffic could be served quickly, in the absence of urgent traffic. Two types of traffic priorities, normal and urgent have been used for performance evaluation of FROG-MAC and DFROG-MAC, over Contiki platform, with the scope of this study focused on vehicle-to-infrastructure (V2I) single-hop communication. The delay and throughput both have been found to improve for DFROG-MAC due to the possibility of dynamic fragment size selection.

1 | Introduction

Vehicular ad hoc networks (VANETs) is the term often used to describe a specialised form of mobile ad hoc networks (MANETs) where vehicles act as network nodes. These networks facilitate vehicle-to-vehicle (V2V) and vehicle-to-infrastructure (V2I) communication, enabling real-time data exchange to enhance traffic safety, optimise transportation efficiency and support applications such as autonomous driving, collision avoidance and emergency services [1]. By dynamically establishing wireless connections, VANETs address the unique challenges of high mobility and frequent topology changes inherent in vehicular environments, making them a cornerstone of intelligent transportation systems (ITS) [2].

Given the high-speed mobility and dynamic topology of VANETs, efficient communication protocols are critical to ensure the reliability and timeliness of data transmission [3].

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Prioritised medium access control (MAC) schemes are particularly essential for VANETs as they help manage the heterogeneous nature of network traffic [4]. In VANET environments, different types of data, such as emergency alerts, safety messages and infotainment data, have varying levels of urgency and quality-of-service (QoS) requirements [5, 6] For example, emergency and safety-critical data, such as collision warnings or vehicle position updates, require immediate transmission with minimal delay to prevent accidents and enhance road safety. On the other hand, noncritical data, such as entertainment or routine updates, can tolerate higher latency [7, 8]. Furthermore, the evolution of VANETs, coupled with the increasing demand for transmitting multimedia content [9] also amplifies the complexity of prioritising critical data.

Without a prioritised MAC scheme, high-priority data may experience delays due to congestion or contention on the communication channel, potentially leading to catastrophic outcomes. Prioritised MAC protocols ensure that urgent messages receive timely access to network resources, thus maintaining the efficiency, reliability, and safety of VANET operations [10]. Some of the most common techniques being used for achieving a prioritised QoS for heterogenous data at the MAC layer include nonoverlapping contention, differentiated channel access, priority queueing multi-channel allocation and hybrid approaches [11]. Among these, IEEE 802.11-p has been widely adopted as the de facto standard for VANETs [12]. It modifies the basic service set (BSS) mechanism to support communication without association or authentication, and employs the utside the context of a BSS (OCB) mode for direct V2V and V2I communication [13]. Prioritisation is achieved through the enhanced distributed channel access (EDCA) scheme, which enables differentiated channel access for varying traffic classes and forms the basis for prioritised QoS in VANETs.

Despite being efficient in prioritising the channel access for urgent data, the major limitation of the above-described approaches is nonavailability of interruption mechanism for the ongoing transmission on the channel. In our previous work, we developed FROG-MAC [14] which offers a quicker chance for higher priority data to transmit by splitting the lower priority data into fragments; the higher priority data get a chance to send the intimation packet during the intervals scheduled between two consecutive fragment transmissions. Due to the unique feature of splitting the ongoing transmission, FROG-MAC showed significant performance improvement for wireless sensors operating in the healthcare [15] and vehicular [8] environments.

The present work enhances FROG-MAC by introducing a yet another protocol dynamic FROG-MAC, where we dynamically adjust the fragment size of lower priority data in the real-time to further reduce the delay for urgent vehicular data. Although FROG-MAC considerably improved the delay performance for urgent data, it still did not dynamically adjust the fragment size of the lower priority data; this limitation may cause serious lacking for the VANET operation where even the delay of msecs cannot be tolerated [16]. Moreover, the scheme could also degrade the network performance significantly for the situations where lower priority data could not transmit quickly even if there is no higher priority traffic is not available.

In this context, we believe that considering the frequency of urgent packet arrival on the network and adjusting the fragment size for lower priority data will further improve the QoS of VANETs. Such approach shall ensure that when urgent traffic is available, it will have a quick chance to transmit, by having interruptible periods in between the ongoing transmission of normal packets; on the other hand, if there is no urgent traffic, the lower priority traffic could use the available bandwidth efficiently. To the best of our knowledge, a scheme, such as DFROG-MAC, has not been proposed in the past.

We list the major contributions of this article below:

1. To propose a dynamic fragmentation-based MAC protocol DFROG-MAC, with the goal of delay reduction and throughput optimisation for VANET.
2. To conduct performance evaluation of DFROG-MAC through performing simulations.
3. To demonstrate the impact of dynamic fragment size adjustment on improving the delay and throughput by comparing DFROG-MAC with FROG-MAC, BoP-MAC and IEEE 802.11-p, which did not have dynamic fragmentation.

Rest of this paper has been organised as follows: Section 2 briefly details the relevant work; Section 3 presents the design of DFROG-MAC; Section 4 details the experimental settings; Section 5 presents the results and analysis and finally and Section 6 concludes the work and offers directions for future research.

2 | Relevant Work

Various dynamic MAC schemes, with a focus on priority assignment and prioritised channel access for vehicular networks have been proposed over the recent past. In this section, we present a brief review of such schemes:

TDMA has been one of the most famous approaches when it comes to assigning priority to the nodes in any wireless network. In such schemes, each node is allocated a specific time slot for transmission, ensuring minimal collisions and efficient utilisation of the shared medium [17, 18]. TDMA provides predictable performance, making it suitable for applications with stringent real-time and QoS requirements, such as industrial IoT, healthcare monitoring systems and VANETs [19]. TDMA has been used both for single and multi-channel vehicular networks [20, 21]. Based on TDMA, self-adaptive time division multiple access (SATMAC) has been proposed in ref. [22] where the allocated time slots for each node are dynamically adjusted based on the information of packet collision of up to 3 neighbours. Similarly, traffic-aware TDMA-based (TA-TDMA) protocol has been proposed [23], where traffic is predicted and slots are accordingly allocated to the vehicles, in order to timely deliver the delay intolerant safety critical messages. Although TDMA offers a dynamic slot allocation for nodes based on the data priority and packet collision, there is still a high delay expected for urgent data for the situations when the lower priority data are being transmitted.

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A distributed cooperative MAC optimisation strategy for VANETs has been proposed in ref. [24], where CSMA has been used in integration with selective roadside unit (RSU) probing and adaptive transmission mechanisms. The scheme dynamically adjusts contention windows for vehicular nodes based on their distance from RSUs. Although this protocol models the dynamic conditions of VANETs, there is no incorporation of multi-priority data generated by these nodes.

Fuzzy logic-based dynamic MAC protocols have also been proposed for VANETs [25]. The focus of these protocols is mostly on adapting transmission parameters dynamically based on varying traffic conditions and network requirements. A Fuzzy based prioritised message dissemination and cooperative MAC (FPMDC-MAC) protocol for VANET has been proposed in ref. [26] The critical warning messages (CWMs) are assigned priority based on their level of severity, whereas the priority of normal packets is determined based on speed, direction and distance between the vehicles using fuzzy models. Moreover, cooperative relay nodes are also used for quick dissemination of CWMs. Despite the efficient priority allocation for CWM, FPMDC-MAC would not be able to allocate channel to the higher priority traffic if noncritical messages have occupied all of the channel. Fuzzy logic has also been used in fuzzy logic-based vehicle weighting model (FWDM) [27], where RSUs are responsible for weighing the vehicles in terms of priority; this decision is made based on the data each RSU receives from the cluster heads (CHs). This approach also tends to exhibit significant complexity and operational overhead, making it less efficient for the highly dynamic and rapidly changing nature of VANET environments.

An interesting multi-channel safety message broadcast scheme has been proposed in ref. [28] where inefficient usage of dedicated short-range communication (DSRC) spectrum has been addressed. The authors propose that if safety message does not find any channel available for transmission, any one of the ongoing nonsafety messages is switched over to the adjacent 5.8 GHz ISM/UNII-3. A similar approach has also been presented in ref. [29], where most efficient channel is selected for the transmission of urgent traffic using the Markovian decision model that takes channel performance, frequency range, power limit and distance as inputs. Although this approach allows transmission of safety messages by removing the ongoing lower priority messages, it requires multi-channel spectrum sharing. Clearly, this would not only require specific hardware but also additional synchronisation messages will also be needed.

Enhancing access probability and reducing access delay (EAPRAD) [30] also uses a Markov model to ensure that all nodes should get a transmission opportunity. The issues of indefinite time slot reservations have been addressed by identifying the node with most failed attempts and subsequently assigning channel for its data transmission. EAPRAD also introduces a new MAC frame structure that uses information of driving directions to partition the available channel resources while also reducing the collisions. EAPRAD has several gaps: first, the protocol attempts to prioritise the nodes instead of data; in case the node which has not been having any transmission opportunity does not have any critical data, there was no need for it to get the channel access. Secondly, there is no mechanism of preempting the ongoing lower priority transmission in case a higher priority packet was to transmit.

IEEE 802.11-p is the standard protocol specifically designed to support vehicular networks. Unlike conventional IEEE 802.11, it is optimised for high-mobility environments by enabling direct communication between vehicles and roadside units through its outside the context of a BSS (OCB) mode [31]. Operating in the 5.9 GHz band, it allocates seven 10 MHz channels, consisting of one control channel (CCH) and six service channels (SCHs). The CCH is dedicated to safety-critical messaging, whereas SCHs carry nonsafety data, such as infotainment and traffic management, thereby allowing 802.11-p to prioritise urgent safety communications while supporting other vehicular applications. Communication delays are further reduced by eliminating association and authentication procedures, which makes the protocol suitable for real-time vehicular scenarios [32]. Prioritisation is achieved through EDCA, which classifies data into four access categories (voice, video, best effort and background), each with specific contention window (CW) sizes and arbitration interframe spaces (AIFS); this ensures that high-priority traffic, such as safety-critical alerts, experiences shorter waiting times and quicker channel access [33]. The use of AIFS for different traffic priorities has been illustrated in Figure 1. Although these mechanisms improve delay performance in moderately loaded networks, a key limitation of 802.11-p remains: once a lower-priority transmission has begun, urgent traffic cannot preempt it, potentially increasing delay during congestion.

The underlying protocol FROG-MAC, on the basis of which DFROG-MAC has been developed in this article, has now been summarised for the reader's interest. FROG-MAC was proposed in ref. [14] with a focus on enabling the higher priority traffic to interrupt the ongoing transmission of lower priority data in a single-hop environment (source-sink communication). The basic operation is illustrated in Figure 2 [14], where after conventional CCA and back-off node-1 sends a RTS. Upon receiving CTS from the sink, node 1 starts to send packet in fragments; however, during one of the pauses between fragments, the sink receives RTS about the urgent packet from another node, node-2. When the sink transmits CTS for node-2, node-1 also listens to it and gets to know about the transmission of urgent traffic; in response, it delays it fragments and continues their transmission after the transmission of urgent packet completes. More details of FROG-MAC can be found in ref. [14].

The other protocol which we have used for performance validation of DFROG-MAC is priority-based medium access control (BoP-MAC) [34]. BoP-MAC uses adaptive back-off mechanism [35] for dealing with heterogenous traffic in IoT and integrates the mechanisms such as duty cycles, active/sleep periods, and adaptive backoff contention. BoP-MAC uses a different contention window for each priority as shown in Figure 3 [34]; this helps the protocol to allocate channel earlier to the higher priority traffic. Simulation results showed that BoP-MAC improved upon existing protocols, such as timeout multi-priority-based MAC (TMPQ-MAC) [36], and is particularly suitable for large-scale networks. However, it is only effective in terms of assigning a lower back-off value for the higher priority traffic. Once the lower priority packet wins the channel and

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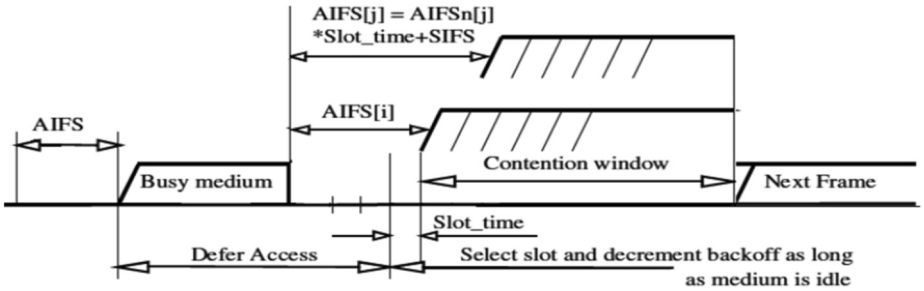

FIGURE 1 | Priority assignments in IEEE 802.11-p [33].

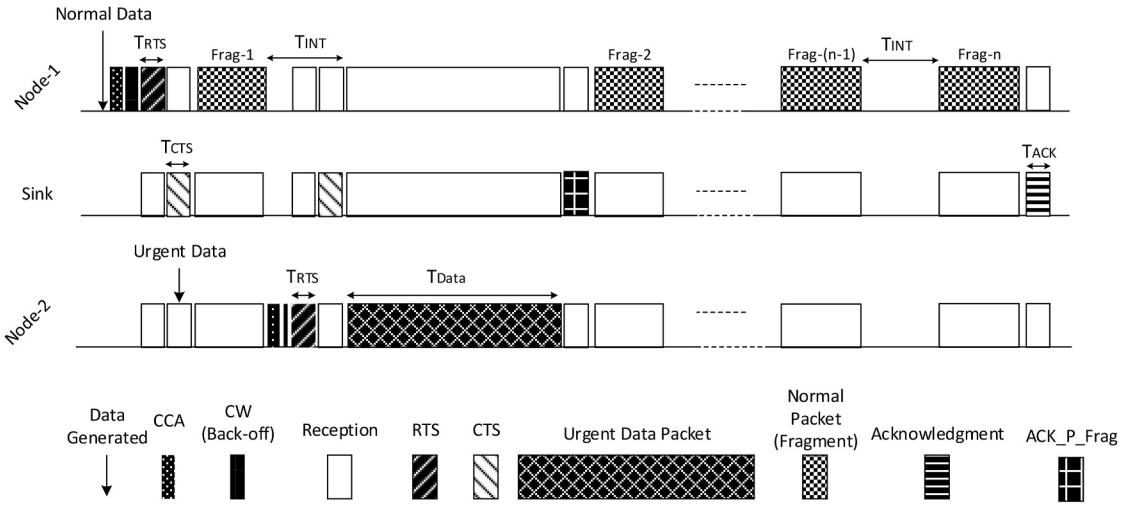

FIGURE 2 | Basic operation of FROG-MAC indicating the channel interruption by urgent traffic.

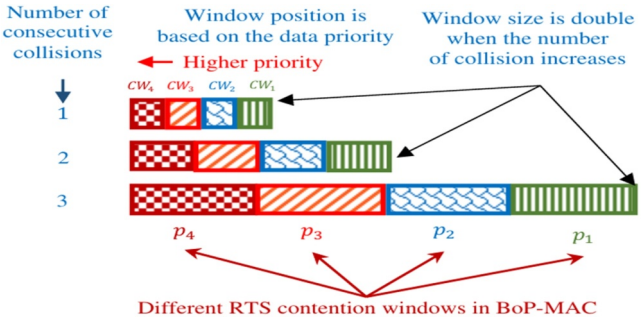

FIGURE 3 | Prioritised contention window access for BoP-MAC.

begins transmission, there is no way using which a higher priority packet could take over and begin an early transmission.

3 | Proposed Design of Dynamic FROG-MAC (DFROG-MAC)

For DFROG-MAC, the major objective is to set the fragment size of low priority/normal packet dynamically in order to optimise the network bandwidth, while still facilitating high priority/urgent traffic with quicker service; this concept has been illustrated in Figure 4.

To make the process of fragment size selection dynamic, the arrival of urgent packet is observed during each period defined as 'Assessment Cycle', T_A. Initially, normal and urgent, both types of packets are not fragmented. As soon as an urgent packet arrives, the normal packet is started to fragment and the duration of T_A is reset. For example, as depicted in Figure 4, an urgent packet arrives during the first cycle of T_A; hence, the normal packet N is fragmented in 2 fragments, N_1 and N_2. Here, after the first occurrence, urgent packet does not arrive again during second cycle of T_A; therefore, the number of fragments decreases until the normal packet returns to its full fragment size N. For the third cycle, no urgent packet arrives and the normal packets continue as full packets. Subsequently, during the fourth and fifth cycles, two urgent packets arrive; thus, the normal packet is fragmented into 4; this way, number of fragments begin to increase until the number of fragments reaches maximum. We also summarise the protocol's operation in Figure 5.

It is to be noted that increase or decrease in number of fragments for DFROG-MAC takes places in steps rather than immediate returning to the original packet size or maximum number of fragments. This stepwise adjustment enhances network stability by preventing abrupt shifts that could destabilise performance under dynamic traffic conditions. It also ensures efficient adaptation to real-time variations in network load, minimising retransmissions and avoiding excessive overhead. By maintaining a gradual change, DFROG-MAC preserves prioritisation for critical data streams, ensuring reliable delivery even during congestion. Furthermore, this approach supports energy efficiency, especially for resource-constrained devices, by avoiding the power spikes associated with sudden changes in

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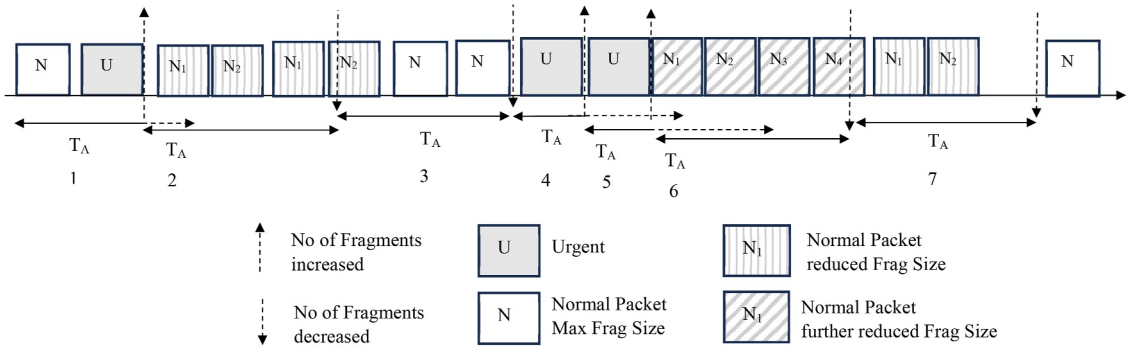

FIGURE 4 | Timing diagram illustrating basic operation of DFROG-MAC.

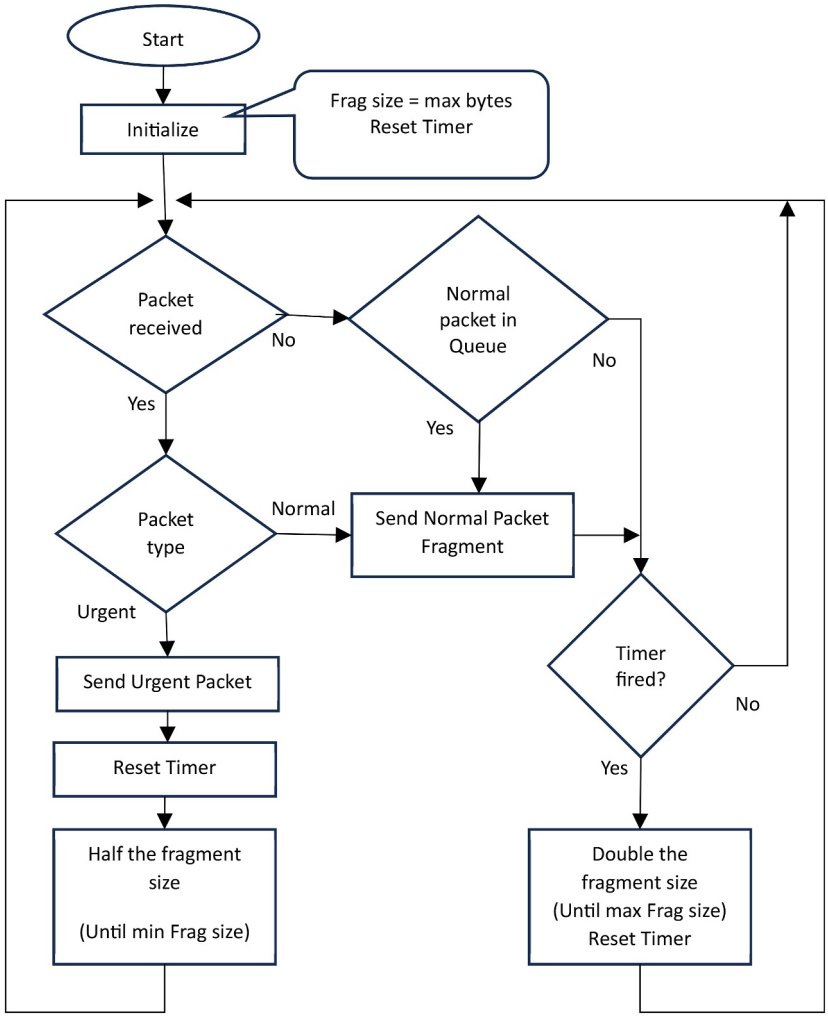

FIGURE 5 | Flow diagram of DFROG-MAC.

fragmentation levels. These benefits collectively contribute to the protocol's robustness and efficiency in managing diverse traffic scenarios.

4 | Experimental Settings

A single-hop star topology network of 11 nodes (10 sources with a single sink) has been used for simulation; this topology exemplifies a basic yet practical vehicular network scenario. In such setups, multiple source nodes (vehicles) communicate directly with a central sink node (e.g., a roadside unit or a central controller) within their coverage area. This topology mirrors real-world vehicle-to-infrastructure (V2I) communication, where vehicles send data, such as traffic updates or sensor readings, to a centralised entity for processing and decision-making, ensuring efficient and streamlined information flow in vehicular environments. Moreover, two levels of priorities have been used to signify normal and urgent data. Here, each

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TABLE 1 | Simulation parameters.

Simulation parameter	Value
Simulation duration	1000 Sec
Contention window size	Non-overlapping
	0 to 10 for urgent traffic
	11 to 20 for normal traffic
Network size	2 to 11
Message generation interval for urgent traffic	2 s (poisson)
Message generation interval for normal traffic	200 msec (CBR)
Data packet size	127 bytes
Total payload length	121 bytes
Fragment size (payload length in each fragment)	Varying (2–121 bytes)
No. of fragments	Varying (1–61)
RTS/CTS packet size	5 bytes
ACK packet size	5 bytes
SACK/NACK packet size	6 bytes
ACK_P_frag size	8 bytes
Time taken to transmit each byte (packetisation delay)	32 μsec
Interruptible period T_{INT}	0.6 ms
Time for clear channel assessment (CCA) T_{CH}	32 μsec

experiment was repeated 5 times and the results have been illustrated with 95% confidence. The details of simulation parameters are listed in Table 1:

5 | Results and Discussions

We conducted performance assessment for the proposed DFROG-MAC by studying the parameters average delay and throughput for urgent and normal priority traffic. The results have been compared with FROG-MAC and BoP-MAC for validation.

5.1 | Delay Analysis

Figure 6 illustrate the performance in terms of average delay when total number of nodes and Assessment timer were varied, respectively; for FROG-MAC and DFROG-MAC, fragment size of 16 was chosen for this experiment. As shown in Figure 6a, the average delay increases for all the protocols with the increasing nodes; this happens because of the increased network congestion, collisions and retransmissions. When comparing for urgent packets, FROG-MAC shows the best performance as it was explicitly designed improving the performance of urgent traffic, without offering any advantage to the normal traffic. Second best contestant for urgent traffic appears to be DFROG-MAC because it still fragments the lower priority packets. On the other hand, the performance of BoP-MAC appears to be lower as compared to both FROG-MAC and DFROG-MAC as it only employs contention window prioritisation, and if the transmission of lower priority packet starts, it will be completed before any higher priority data could transmit. In comparison, IEEE 802.11-p exhibits significantly higher average delay for urgent packets, primarily due to its contention-based channel access mechanism, which does not prioritise emergency traffic effectively.

For the normal traffic, BoP-MAC performs better as once the channel will be granted to the normal traffic, the packets will be transmitted regardless of the availability of higher priority data on the network. On the other hand, FROG-MAC appears to having the highest average delay for normal traffic as it does not adaptively adjust the fragment size, and normal packets were sent always in fragments. Furthermore, IEEE 802.11-p also shows comparatively higher delay for normal traffic as the lack of fragmentation and adaptive prioritisation mechanisms leads to increased waiting times under heavy network load. DFROG-MAC has proven to reduce the delay as it offers much higher chance of normal packets to transmit earlier, in case there are no higher priority packets available on the network.

Figure 6b shows that with the increasing value of Assessment Timer (T_A), the average delay for urgent traffic reduces, whereas it increases for the normal traffic. This trend can be explained with the help of Figure 4, where it was shown that in case a longer T_A is selected, the lower priority packets will be transmitted in smaller fragments for long period, which leads to a higher delay for low priority. On the other hand, shorter fragments of lower priority traffic indicates that urgent data could be transmit earlier as they will get a higher probability for channel interruption.

We subsequently conducted experiments for fragment size 2 for FROG-MAC and the results for average delay are shown in Figure 7. As compared to Figure 6a, the values of delay for FROG-MAC increase for normal and reduce for urgent traffic in Figure 7, due to the reduction in fragment size. This happens again because due to being split in more fragments, a lower priority packet will face a higher delay, whereas urgent packets will get a quicker chance to transmit. On the other hand, since fragment size is dynamic for DFROG-MAC and it has no relevance for 802.11-p and BoP-MAC, there is no difference in delay results for these protocols.

5.2 | Throughput Analysis

Figure 8 compares the throughput of 3 protocols for urgent and normal traffic; here, Figure 8a illustrates the case of fragment size 16, whereas Figure 8b shows the results for fragment size 2. As seen from both these figures, up to 4 nodes, the throughput of all protocols remained same, approximately 100%. As the traffic load is low and there are lower collisions, the channel usage comes out to be best at the low loads. Subsequently, as the

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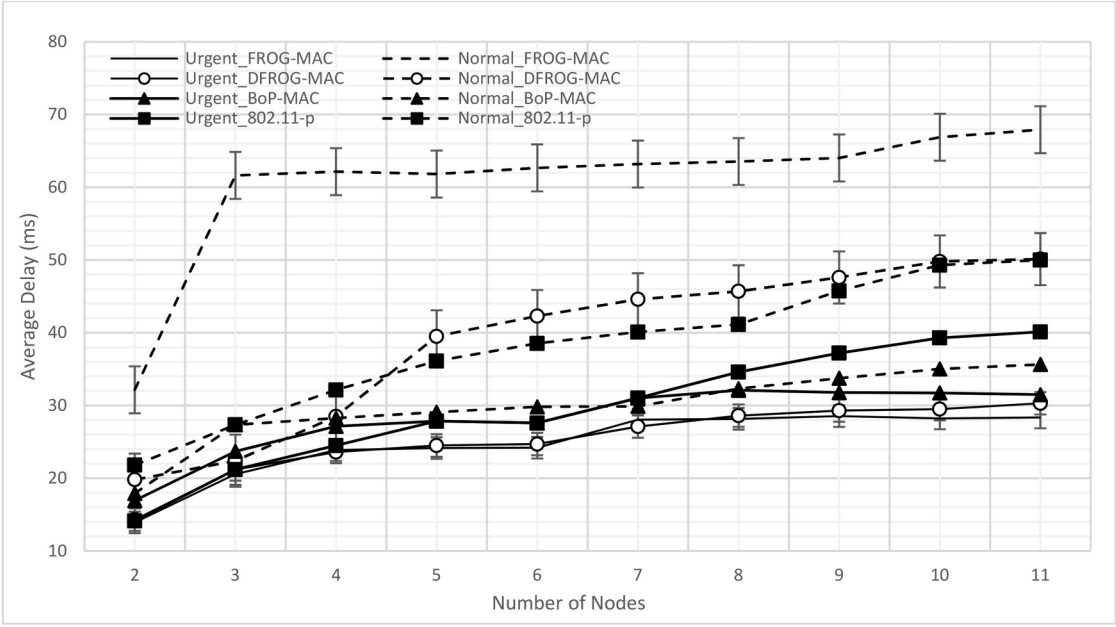
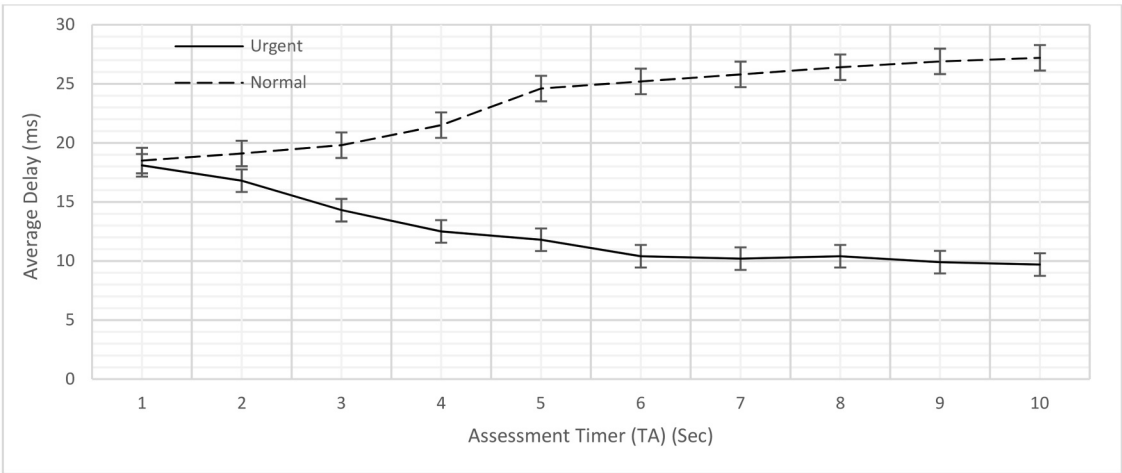


(a)



(b)

FIGURE 6 | Average delay for DFROG-MAC in comparison to FROG-MAC and BoP-MAC (fragment size = 16) (a) varying number of nodes and (b) varying assessment timer.

load increases on network, the throughput reduces for all the protocols, with the most decline for BoP-MAC for normal traffic. Again, since BoP-MAC does not employ the fragmentation, the normal traffic will get the lowest channel access. IEEE 802.11-p shows a comparatively sharper drop in throughput as the number of nodes increases for both urgent and normal traffic. This is mainly because its contention-based access without fragmentation or priority adaptation results in higher collision probability and reduced channel efficiency under heavier loads. On the other hand, the best protocol in terms of throughput comes out as FROG-MAC for urgent traffic, whereas the second-best performer is DFROG-MAC. From the perspective of normal traffic, DFROG-MAC appears to be the best as it allows the normal traffic to also share the channel efficiently for the instants when higher priority data is not available on the channel. These trends remain same also for the fragment size 2 as shown in Figure 8b.

5.3 | Computational Complexity Analysis

To assess the practicality of DFROG-MAC on resource-constrained VANET devices, we analysed its computational overhead with reference to Contiki motes, such as *Tmote Sky/TelosB*, which are equipped with MSP430 microcontrollers (8 MHz, 10 KB RAM, 48 KB Flash and CC2420 radio).

- *Processing Requirements*: The dynamic fragmentation logic consists of counter increments, simple comparisons and conditional halving/doubling of fragment sizes. On the MSP430, this requires only tens to a few hundred CPU cycles corresponding to ≈10–20 μs at 8 MHz. In contrast, radio transmission dominates the cost; transmitting a 10-byte fragment at 250 kbps requires ≈320 μs. Thus, the compute overhead per decision is negligible relative to the airtime of even the smallest fragment.

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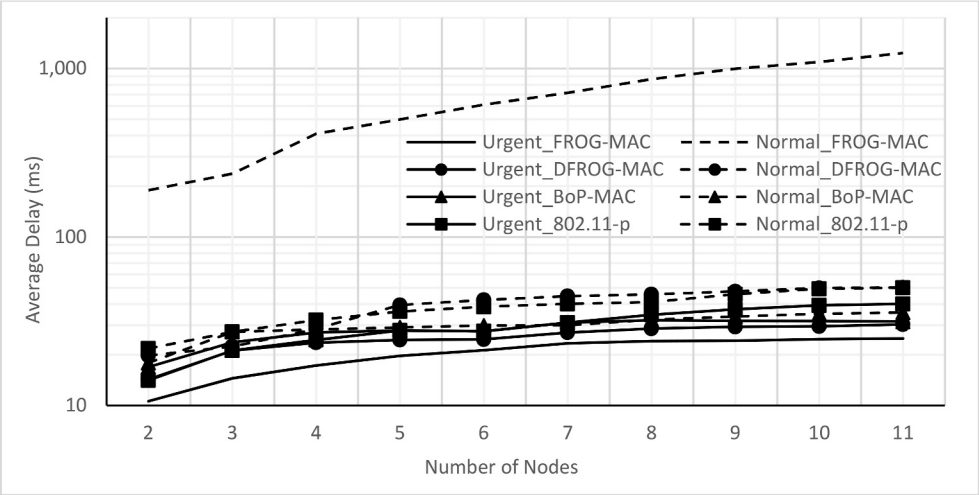
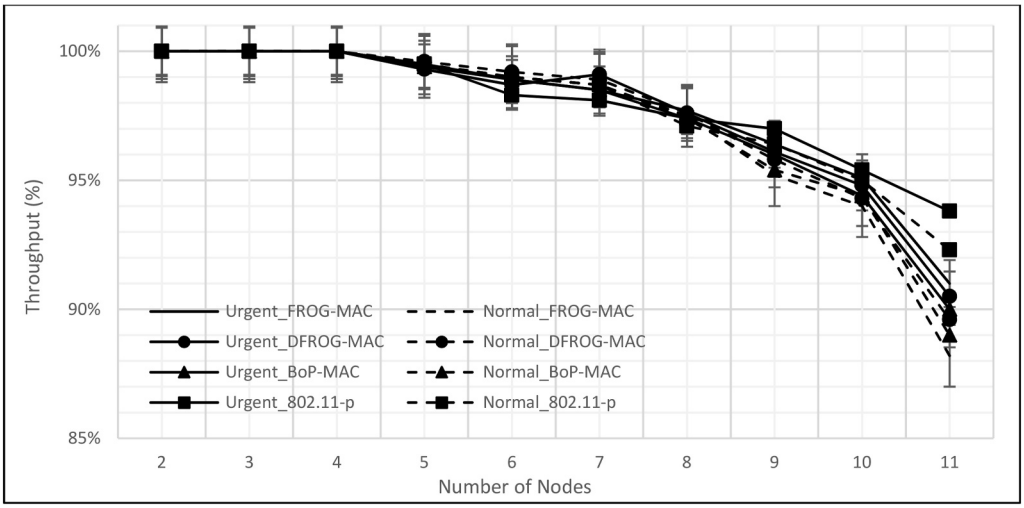
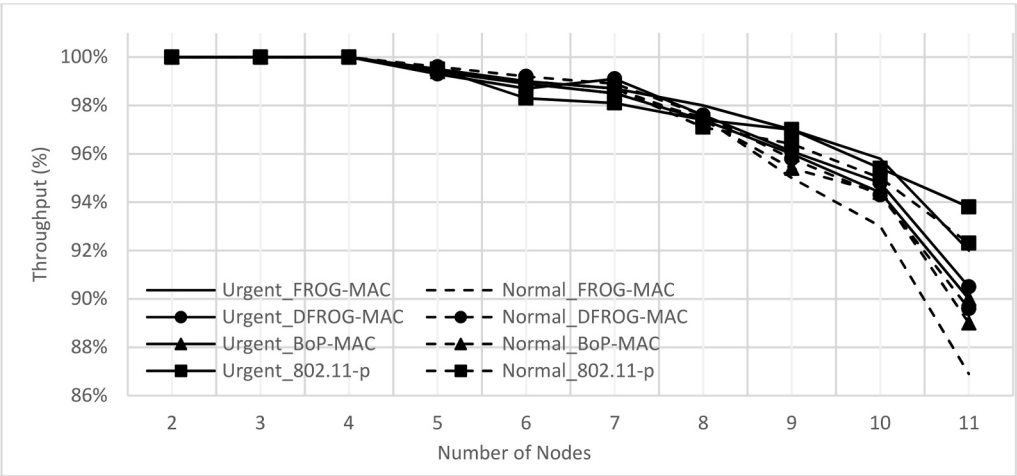


FIGURE 7 | Average delay for DFROG-MAC in comparison to FROG-MAC and BoP-MAC (fragment size = 2).



(a)



(b)

FIGURE 8 | Throughput performance for DFROG-MAC in comparison to FROG-MAC and BoP-MAC: (a) fragment size = 16 and (b) fragment size = 2.

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- *Memory Footprint*: DFROG-MAC requires additional state variables for fragment size, bounds, urgent arrival counters and timer references. The total memory cost is < 20 bytes of RAM, which is trivial compared to the 10 KB available on TelosB/Tmote Sky motes.
- *Energy Budget*: The MSP430 consumes ≈5.4 mW in active mode. A fragmentation decision taking 20 μs costs ≈0.1 μJ, which is negligible compared to the radio cost of transmitting even a 10-byte fragment (≈24 μJ, given the CC2420 draws ≈20 mA @ 3 V). It is acknowledged that fragmentation increases the number of transmission events and thus incurs additional radio energy due to extra headers and acknowledgements. However, DFROG-MAC offsets this overhead by allowing urgent packets to interrupt ongoing transmissions, thereby reducing retransmissions caused by missed deadlines or collisions. In practice, avoiding repeated retransmissions of urgent packets saves more radio energy than the extra fragments consume. Hence, although fragmentation increases per-packet overhead, the overall system energy efficiency is preserved because critical traffic is delivered reliably without costly retransmissions.
- *System Impact*: Because the algorithm is O(1) per packet and does not require extra control messages, overhead remains constant per node and scales linearly with the number of active motes. The dominant energy and delay factors continue to be radio contention and retransmissions, not computation.

In summary, the computational overhead of DFROG-MAC remains minimal, confirming its feasibility for deployment on Contiki motes in VANET environments.

6 | Limitations

Although DFROG-MAC demonstrates promising results for prioritised data management, several limitations should be acknowledged. First, the current design and evaluation have been conducted under a single-hop topology, reflecting V2I communication. However, many VANET scenarios rely on multi-hop forwarding, where cumulative delay, synchronisation between relays and fairness among forwarding nodes become critical. These aspects remain outside the present scope and will be investigated in future extensions of the protocol.

Another limitation of the present study is that the Assessment Timer (T_A) was kept constant for all experiments. Although this provides a controlled baseline for evaluating the impact of dynamic fragmentation, it may not fully capture the highly dynamic traffic load variations typical in VANET environments. In practice, an adaptive T_A mechanism, responsive to real-time traffic density, could further optimise performance. This remains outside the scope of the current work but is identified as an important direction for future research.

Furthermore, the simulations assume idealised channel and traffic models, whereas real vehicular environments are affected by shadowing, fading and bursty traffic conditions. Incorporating these variations will provide a more comprehensive performance assessment. Finally, the current evaluation considers only two priority classes (urgent and normal). Although this is sufficient to demonstrate the principle of dynamic fragmentation, future work should explore finer-grained priority levels to address the diverse traffic demands in modern VANETs such as safety-critical alerts, infotainment and control signalling.

7 | Conclusion and Future Work

This paper presented a dynamic fragmentation-based MAC protocol DFROG-MAC for facilitating quick transmission of high priority packets for mission critical applications such as vehicular networks. The lower priority packets are transmitted in fragments to let the higher priority packet interrupt the transmission. However, DFROG-MAC also accounts for challenging situations where higher priority data may not be available on the channel, lower priority traffic should get an opportunity for transmission. Hence, the fragment size for the lower priority packet is determined by analysing the packet arrival rates for the higher priority packets; this allows the node to optimise the usage of channel resource, while reducing the delay for packets. The performance of DFROG-MAC has been shown to be better than the earlier variants FROG-MAC, BoP-MAC and IEEE 802.11-p in terms of delay and throughput. Specifically, DFROG-MAC is expected to enhance the efficiency of VANETs for multimedia transmissions.

In future, this work may be extended in several directions. One avenue is developing multi-hop variants of DFROG-MAC, which could be evaluated using existing Contiki multi-hop testbeds to assess scalability. Another direction is the integration of machine learning or fuzzy logic to optimise fragment sizes based on real-time traffic patterns. For instance, lightweight models (e.g., decision trees or fuzzy rule sets) can be implemented on motes to classify traffic load and adapt fragmentation without excessive computation. Such models have been successfully deployed on MSP430-class hardware in prior Contiki studies, suggesting feasibility for DFROG-MAC. Additionally, energy-aware scheduling could be coupled with QoS adaptation to improve multimedia transmissions, whereas experimental validation in real vehicular testbeds would further enhance credibility. Finally, adapting DFROG-MAC for heterogeneous networks and incorporating adaptive scheduling for varying traffic loads remain promising directions.

Author Contributions

Anwar Ahmed Khan: conceptualization, software. **Shama Siddiqui:** conceptualization, methodology, writing – original draft. **Ahmad Sami Al-Shamayleh:** writing – review and editing. **Adnan Akhunzada:** writing – review and editing. **Indrakshi Dey:** project administration, resources.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The datasets used and/or analysed during the current study available from the corresponding author on reasonable request.

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